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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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ITA0521-Computer Vision

Computer Vision in Agriculture

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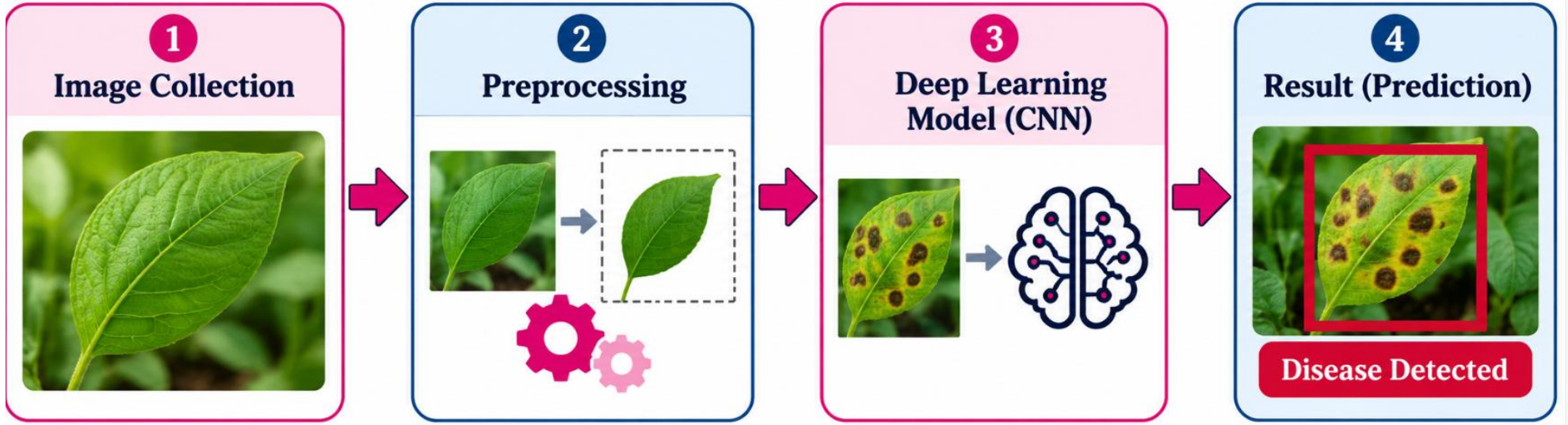
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INTRODUCTION

- Agriculture is important for food production and the economy.
- Farmers need to identify plant diseases, weeds, and crop problems early.
- Computer vision allows computers to understand images and videos.
- It uses cameras and AI to examine crops and help farmers make decisions.
- It can support faster, more accurate, and more efficient farming.



HOW DOES IT WORK?



1

Capture Image

Phone, camera, drone

2

Process Image

Clean and resize

3

Analyze with AI

Detect disease or weeds

4

Show Result

Alert or recommendation

ADVANTAGES

- Early disease detection can help reduce crop damage.
- Saves time by checking many plants or large fields quickly.
- Can identify weeds and support targeted weed control.
- Helps monitor crop growth and estimate fruit or plant counts.
- Supports data-based decisions and more efficient use of resources.

REAL-WORLD APPLICATIONS

- Plant disease detection: recognizes visible symptoms on leaves.
- Weed detection: distinguishes weeds from crops for targeted spraying.
- Fruit counting: estimates the number of fruits before harvesting.
- Crop monitoring: observes growth, color, and signs of stress.
- Drone and satellite imaging: helps inspect large farms and identify problem areas.

FUTURE SCOPE

- More accurate AI models for different crops and local conditions.
- Smartphone apps that provide quick crop health checks.
- Robots and drones that monitor fields with less manual work.
- Combining computer vision with sensors and weather data.
- Precision farming that reduces waste and supports sustainable agriculture.



CONCLUSION

- Computer vision helps computers understand crop images.
- It can detect diseases, identify weeds, count fruits, and monitor growth.
- It can save time and help farmers make better decisions.
- Image quality, cost, and model accuracy remain important challenges.
- With continued improvement, it can play a valuable role in smart farming.

THANK YOU